

STUDENT-TO-TUTOR SCHEDULING AND AVAILABILITY MANAGEMENT SYSTEM

CONCEPT DOCUMENT (MILESTONE 1)

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1. PROBLEM STATEMENT

Students who need tutoring support often do not have a centralized way to see which tutors are qualified for a course, when those tutors are available, or whether a requested meeting fits within approved scheduling rules. When scheduling is handled informally, students may rely on scattered messages, incomplete availability information, or last-minute coordination, which increases the risk of confusion, missed opportunities, and inconsistent appointment records.

Tutors and administrators also face difficulties when there is no shared system for account approval, availability management, appointment tracking, review moderation, and holiday or time-window enforcement. As a result, students may struggle to get timely academic support, while tutors and administrators spend additional time manually answering availability questions, resolving conflicts, and checking whether appointments fall within allowed hours.

2. MISSION STATEMENT

The mission of this project is to design a secure, role-based scheduling system that allows approved students to search for tutors by course and capability, view valid availability, schedule tutoring sessions within approved time rules, and manage reviews after completed meetings. At the same time, the system will provide tutors, administrators, and the super admin with structured controls over availability, schedule rules, profile visibility, basic notifications, and simple reporting so that tutoring operations become more organized, visible, and dependable.

3. PROJECT DESCRIPTION

This Student-to-Tutor Scheduling and Availability Management System is designed for a school environment in which users apply for accounts, are approved by an administrator, and then use role-specific functions to schedule and manage tutoring sessions.

The system is centered on four primary user roles (Student, Tutor, Admin, and Super Admin) plus an Applicant state for users who have not yet been approved. Each role is granted a limited, purposeful set of permissions to protect privacy and reduce scheduling errors.

3.1 BUSINESS PURPOSE

This project addresses two connected academic-support problems: (1) students do not always know when qualified tutors are available, and (2) tutors and staff do not always have a consistent scheduling method that records what has been requested, approved, rescheduled, canceled, or completed. A dedicated scheduling system improves visibility, reduces manual coordination, and makes tutoring services easier to access and manage.

3.2 CONCEPTUAL SOLUTION OVERVIEW

At a high level, the concept includes the following capabilities (described as system behavior, not user-interface layout or technical implementation):

- Account application and approval workflow for new users, with Admin approval or denial before normal access is granted.
- Role-based access control and profile visibility so each user sees only the information necessary for their responsibilities.
- Student ability to store current courses and search tutors by course and capability, with basic filters such as rating and available time.
- Tutor ability to define availability windows for tutoring sessions (subject to global schedule rules and holiday blocks).
- Appointment creation and lifecycle tracking using statuses: Scheduled, Rescheduled, Canceled, and Completed.
- Student-controlled reviews after completed sessions; tutors can view review results but do not control review content.
- Administrative oversight for approvals, profile management, schedule rules, review moderation, and activity totals (simple reporting).
- Conceptual notifications for major workflow events (approval, booking, cancellation, reschedule).

3.3 STAKEHOLDERS, USER ROLES, AND CAPABILITIES

The roles and capabilities below are derived from the project requirements list provided for this course.

Student capabilities:

- Apply for an account and maintain personal profile data (e.g., name, ID, contact details, GPA, year in school).
- Input current courses.
- Search tutors and tutored courses; filter results by course, capability, rating, and available time.
- View only a tutor's basic profile information needed to make a booking decision.
- Select a tutor and schedule a meeting only within valid available times.
- View their own scheduled meeting count and upcoming appointments.
- Create, edit, or remove their own reviews after completed sessions.

Student restrictions:

- Cannot set tutor availability.
- Cannot schedule outside available times, outside approved hours, or on holidays.
- Cannot view full tutor profiles or other users' profiles.

Tutor capabilities:

- Maintain tutor profile data and supported capabilities/courses.

- Set available dates and times (availability windows) for tutoring sessions, within system rules.
- View upcoming appointments and meeting counts.
- Reschedule, cancel, or mark appointments as completed.
- View review results and review counts for their profile.

Tutor restrictions:

- Cannot set availability on holidays or outside designated hours.
- Cannot edit, create, or delete reviews; cannot change their own rating.
- Cannot see more than the basic information on student profiles, and cannot see the student's name within a review.

Admin capabilities:

- Approve or deny account requests; create and delete accounts as needed.
- View and edit full student and tutor profile information.
- Set system-wide schedule rules (allowed time windows) and holiday blocks.
- View and cancel scheduled appointments when required.
- Remove submitted reviews that violate policy.

Admin restrictions:

- Cannot create or edit reviews.
- Cannot edit another admin profile/role (policy constraint).

Super Admin capabilities:

- See all accounts and edit all profiles (highest visibility level).
- Create and delete accounts.
- Set system-wide schedule rules and holiday blocks.
- View system-wide totals and reporting summaries.

Super Admin restrictions:

- Cannot set schedules for tutors (availability is tutor-controlled).
- Cannot book tutoring appointments.
- Only one Super Admin role exists in the system at a time.

Role Capability Summary (Matrix)

Capability	Student	Tutor	Admin	Super Admin
Apply for account / approve access	Apply	-	Approve/deny	Oversee
Maintain own profile	Yes	Yes	Yes	Yes
View other profiles	Basic tutor	Basic student	Full	Full

	only	only		
Search tutors & courses	Yes	-	-	-
Set availability	-	Yes	Configure rules only	Configure rules only
Book session	Yes	-	Can set/cancel	-
Manage appointment status	-	Yes	Cancel	-
Reviews (create/edit/remove)	Yes	View only	Moderate/remove	Moderate/remove
Reporting / totals	Own counts	Own counts	System totals	System totals

3.4 KEY BUSINESS RULES AND CONSTRAINTS

The following rules define the conceptual behavior of the system and are referenced in the use cases:

- Role-based access: Students see only basic tutor profiles; tutors see only the student booking details they need; admins have broader visibility; the super admin has the highest visibility.
- Scheduling windows: Tutor availability and student bookings must be within globally allowed hours and must avoid holiday blocks.
- Appointment lifecycle: Each appointment must have a status (Scheduled, Rescheduled, Canceled, Completed) so that tracking and reporting are consistent.
- Conflict prevention: A time slot cannot be booked if it overlaps an existing scheduled appointment.
- Review privacy: Reviews are created/edited/removed only by the student; tutors can view review results but cannot identify the student from the review display.
- Administrative enforcement: Admins can remove inappropriate reviews and cancel appointments when needed.

3.5 CONCEPTUAL DATA (HIGH-LEVEL)

This section describes the main information the system must manage:

- User account: role (Student/Tutor/Admin/Super Admin), approval status, and core identity fields.
- Student profile: courses currently taken, academic details (as permitted), and meeting history counts.
- Tutor profile: supported courses/capabilities, ratings summary, and review counts.
- Availability window: tutor-defined open time blocks (validated against global schedule rules).
- Appointment: student, tutor, date/time, status, and optional reason notes for reschedule/cancel.

- Review: student-authored feedback linked to a completed appointment and a tutor profile (with privacy controls).
- Schedule rules: global allowed hours and holiday block dates/times.

3.7 HIGH-LEVEL WORKFLOW DIAGRAM

Figure 1 summarizes the end-to-end workflow at a conceptual level.

High-Level Workflow (Concept)

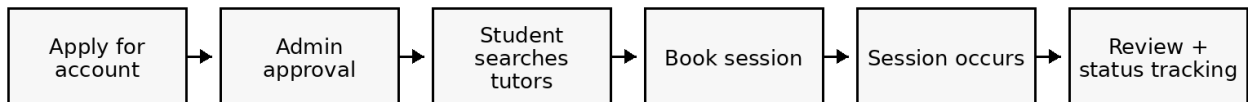


Figure 1. High-level workflow (conceptual).

4. USE CASES

4.1 USE CASE SUMMARY TABLE

ID	Actor(s)	Use Case	Expected Outcome
UC-1	Applicant / Admin	Account application and approval	Only approved users gain normal system access; pending totals update.
UC-2	Student	Search tutors and view basic tutor profiles	Student receives a narrowed list of tutors and their valid available times.
UC-3	Student (primary), Tutor (notified)	Book a tutoring meeting	Appointment is created as Scheduled; confirmation is generated.
UC-4	Tutor	Set availability and manage appointment status	Availability updates; status changes are tracked and communicated.
UC-5	Student (primary), Tutor (viewer)	Create/edit/remove reviews after sessions	Reviews remain student-controlled; tutor sees results without student identity.
UC-6	Admin / Super Admin	Administrative oversight and reporting	System policies are enforced and activity totals are available.

4.2 OVERALL USE CASE DIAGRAM

Figure 2 shows the overall set of use cases (UC-1 to UC-6) and the actors who participate in each.

Overall Use Case Diagram (UC-1 to UC-6)

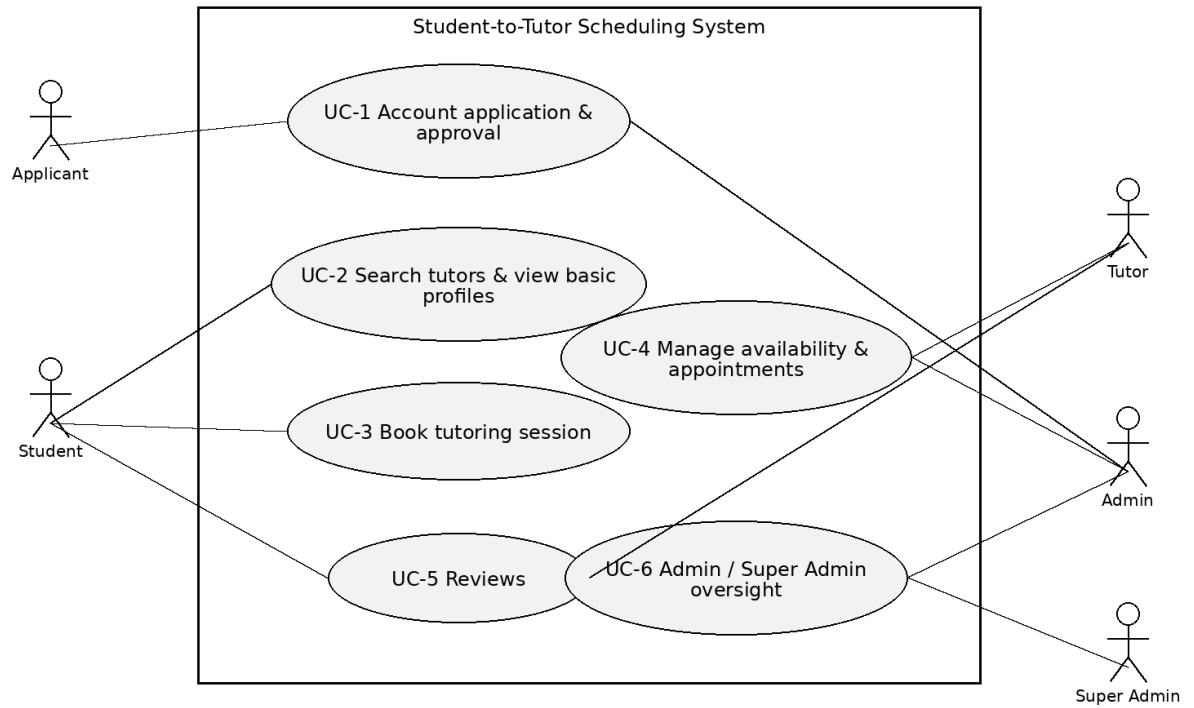


Figure 2. Overall use case diagram (UC-1 to UC-6).

4.3 USE CASE SPECIFICATIONS (UC-1 TO UC-6)

UC-1: Account Application and Approval

UC-1 Use Case Diagram: Account Application and Approval

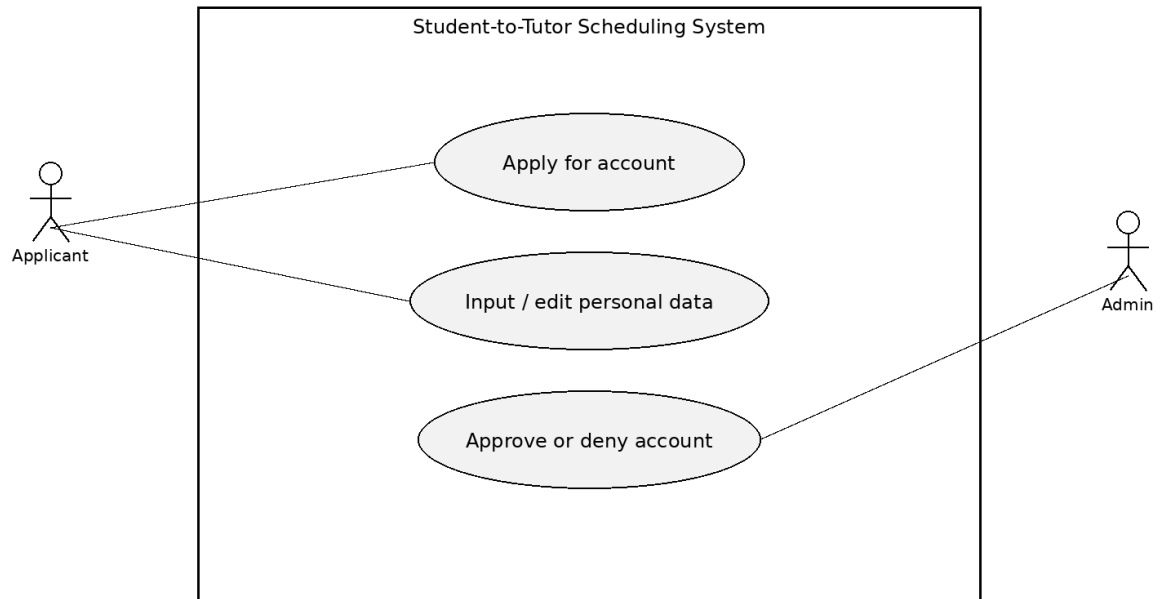


Figure 3. UC-1 use case diagram.

Use Case ID / Name	UC-1 - Account Application and Approval
Primary Actor(s)	Applicant; Admin
Goal	Allow a new user to request access and an Admin to approve or deny access before normal use begins.
Preconditions	Applicant is not yet an approved active user.
Trigger	Applicant submits an account request.
Main Success Scenario	1. Applicant completes the account application with required personal data. 2. System validates required fields and stores the request with status 'Pending Approval'. 3. System notifies Admin of a pending request. 4. Admin reviews the request and either approves or denies it. 5. If approved, system activates the account with the correct role and sends a confirmation notice; if denied, system records the denial and optionally provides a reason.
Alternative / Exception Flows	A1. Missing or invalid required data: system prompts the applicant to correct and resubmit. A2. Admin denies the request: the account remains inactive; applicant is notified of denial (and reason if provided). A3. Applicant withdraws request before approval: system marks request as withdrawn.

Postconditions	Account is either active (approved) or inactive (denied/withdrawn), and pending totals reflect the new state.
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UC-2: Student Search and Tutor Selection

UC-2 Use Case Diagram: Student Search and Tutor Selection

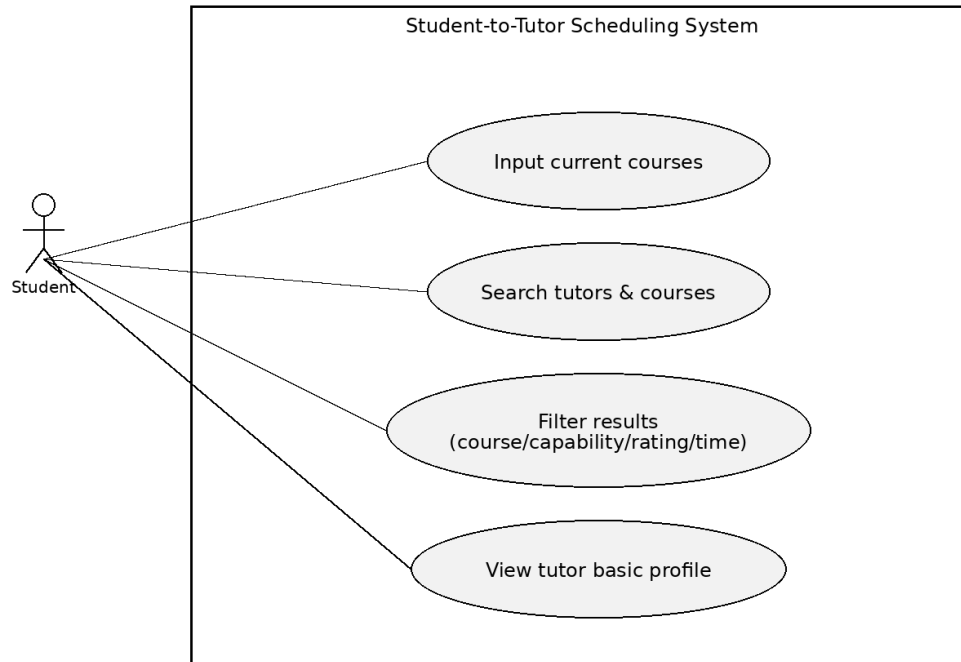


Figure 4. UC-2 use case diagram.

Use Case ID / Name	UC-2 - Student Search and Tutor Selection
Primary Actor(s)	Student
Goal	Help the student find appropriate tutors for a course and view basic tutor profile information needed for scheduling.
Preconditions	Student account is approved and the student is authenticated.
Trigger	Student initiates a tutor search.
Main Success Scenario	1. Student inputs or updates current courses (if not already stored). 2. Student searches tutors and tutored courses. 3. Student applies filters (course, capability, rating, available time) to narrow results. 4. System returns matching tutors and their available times. 5. Student opens a tutor's basic profile to review key details (e.g., capabilities and rating summary).
Alternative / Exception Flows	A1. No matches: system shows an empty result state and suggests adjusting filters. A2. Student has no courses entered: system prompts to add

	courses or search by course keyword.
Postconditions	Student has a list of candidate tutors and can proceed to booking.

UC-3: Student Books a Tutoring Meeting

UC-3 Use Case Diagram: Student Books a Tutoring Meeting

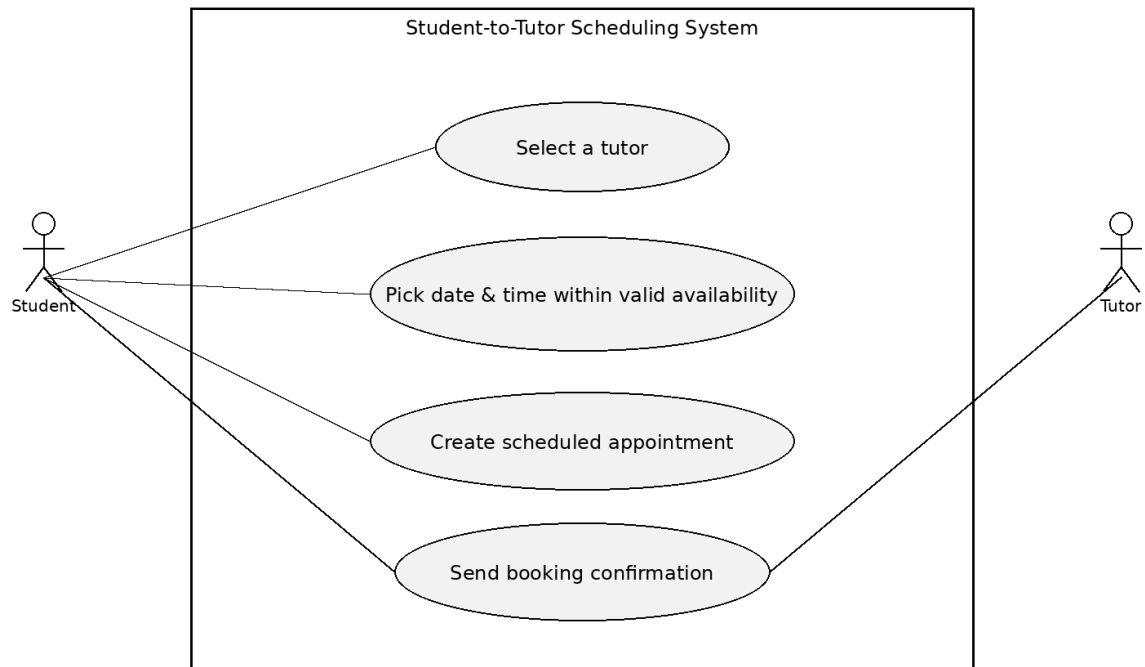


Figure 5. UC-3 use case diagram.

Use Case ID / Name	UC-3 - Student Books a Tutoring Meeting
Primary Actor(s)	Student
Supporting Actor(s)	Tutor (receives booking notification)
Goal	Create a tutoring appointment with a valid tutor, date, and time.
Preconditions	Student is approved and logged in; selected tutor has defined availability; requested time is within global rules and not on a holiday.
Trigger	Student selects a tutor/time and confirms booking.
Main Success Scenario	1. Student selects a tutor from search results. 2. Student chooses an available date and time that fits tutor availability and global schedule rules. 3. System validates the time slot (not on a holiday, within allowed hours, and not already booked). 4. System creates an appointment with status 'Scheduled'. 5. System sends a booking confirmation notification to the

	student and tutor.
Alternative / Exception Flows	A1. Time slot becomes unavailable before confirmation: system rejects booking and prompts the student to choose another time. A2. Requested time violates schedule rules (holiday/outside allowed hours): system rejects booking and explains the constraint.
Postconditions	Appointment exists with status 'Scheduled' and is visible to the student and tutor in their appointment lists.

UC-4: Tutor Availability and Appointment Control

UC-4 Use Case Diagram: Tutor Availability and Appointment Control

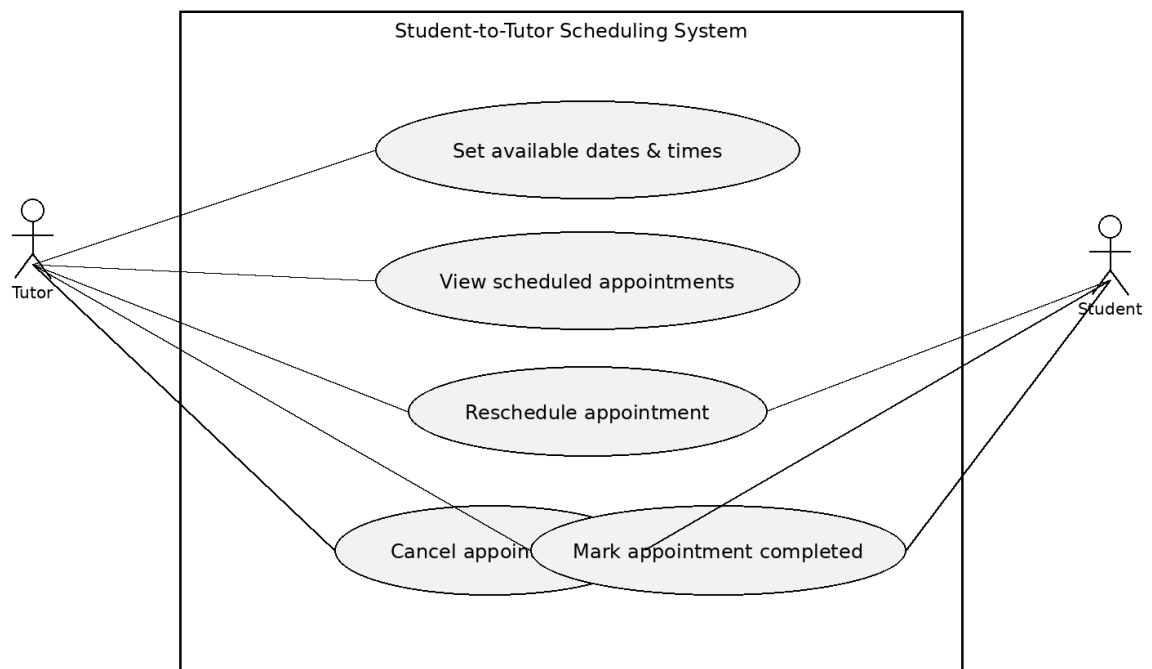


Figure 6. UC-4 use case diagram.

Use Case ID / Name	UC-4 - Tutor Availability and Appointment Control
Primary Actor(s)	Tutor
Supporting Actor(s)	Student (receives changes/notifications)
Goal	Allow the tutor to set availability and manage appointment status changes.
Preconditions	Tutor is approved and authenticated.
Trigger	Tutor updates availability or modifies an appointment.
Main Success Scenario	1. Tutor sets or updates available dates and times (availability windows). 2. System validates availability against global schedule rules and

	holiday blocks. 3. Tutor views scheduled appointments. 4. Tutor reschedules an appointment when needed; system updates status to 'Rescheduled' and notifies the student. 5. Tutor cancels an appointment when needed; system updates status to 'Canceled' and notifies the student. 6. After a session occurs, tutor marks the appointment as 'Completed' so review and reporting can proceed.
Alternative / Exception Flows	A1. Tutor attempts to create availability outside allowed hours or on a holiday: system rejects and explains the rule. A2. Reschedule conflicts with an existing booking: system rejects and prompts for a different time.
Postconditions	Availability windows and appointment statuses accurately reflect the tutor's schedule and are visible to the student and admin roles as permitted.

UC-5: Reviews After Completed Sessions

UC-5 Use Case Diagram: Reviews After Completed Sessions

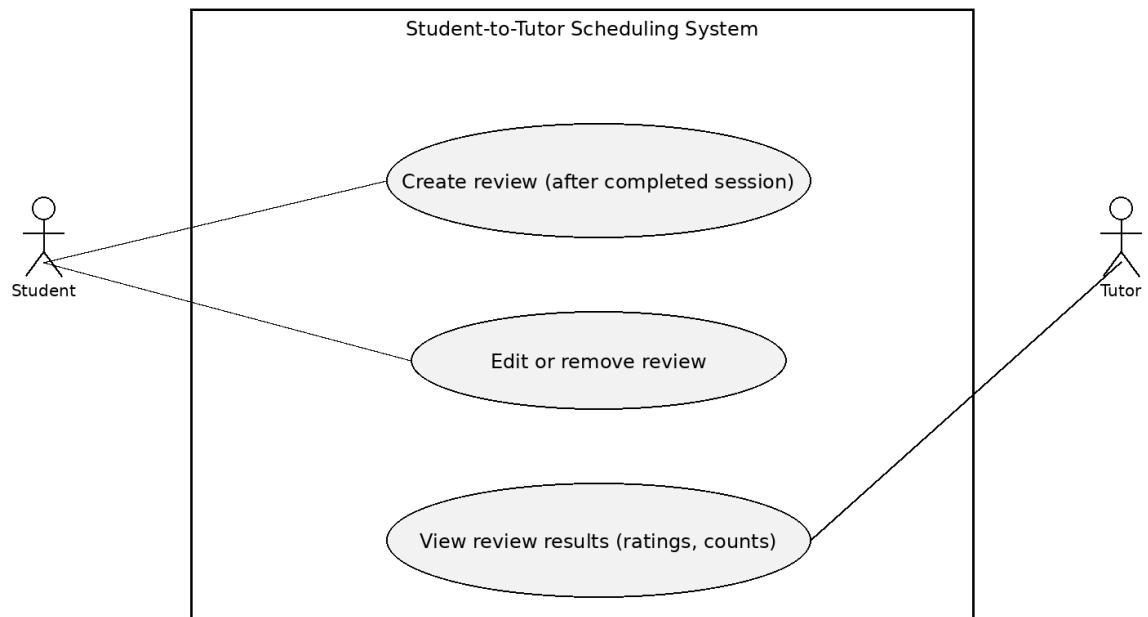


Figure 7. UC-5 use case diagram.

Use Case ID / Name	UC-5 - Reviews After Completed Sessions
Primary Actor(s)	Student
Supporting Actor(s)	Tutor (views results)
Goal	Allow students to leave feedback after completed tutoring sessions while preserving student privacy.
Preconditions	An appointment between the student and tutor has status

	'Completed'.
Trigger	Student chooses to create or modify a review for a completed session.
Main Success Scenario	1. Student creates a review linked to a completed appointment and the tutor profile. 2. System saves the review and updates tutor rating summary and review counts. 3. Student may later edit or remove their own review (policy controlled). 4. Tutor views review results and counts; review display does not reveal student identity.
Alternative / Exception Flows	A1. Student attempts to review before completion: system prevents review creation and explains the precondition. A2. Student removes a review: system recalculates summary counts as needed.
Postconditions	Review is stored (or updated/removed) and tutor-facing summaries reflect the change while keeping student identity private in review display.

UC-6: Administrative Oversight and Reporting

UC-6 Use Case Diagram: Administrative Oversight and Reporting

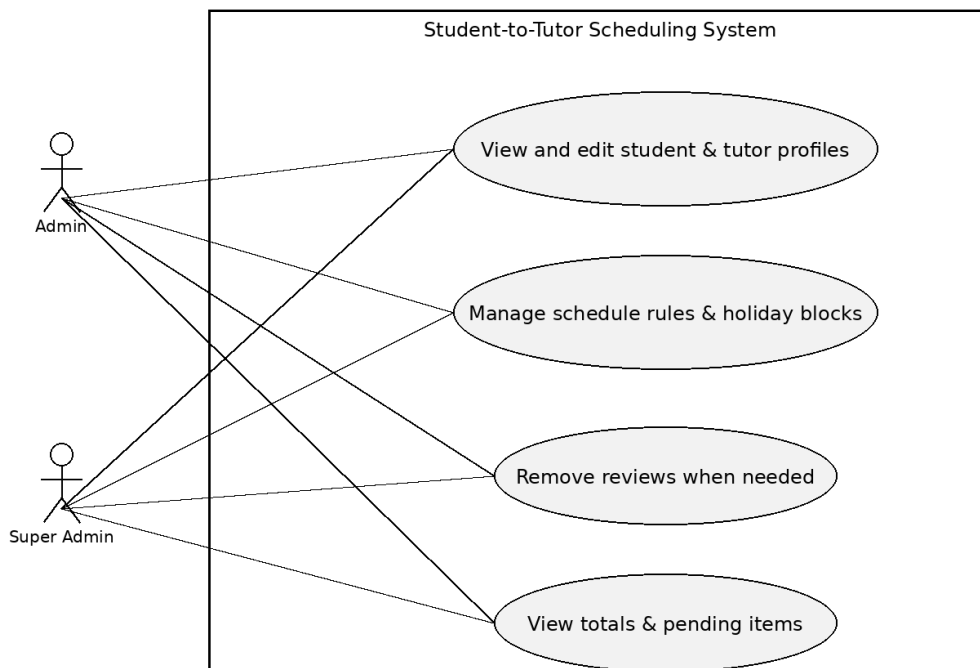


Figure 8. UC-6 use case diagram.

Use Case ID / Name	UC-6 - Administrative Oversight and Reporting
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Primary Actor(s)	Admin; Super Admin
Goal	Support governance of accounts, profiles, schedule rules, and moderation, with visibility into totals and pending items.
Preconditions	Admin or Super Admin is authenticated.
Trigger	Admin/Super Admin performs an oversight action.
Main Success Scenario	1. Admin reviews and manages user accounts (approve/deny/create/delete as permitted). 2. Admin and Super Admin view and edit student/tutor profiles to correct or update information. 3. Admin and Super Admin define global schedule rules and holiday blocks. 4. Admin removes inappropriate reviews when needed. 5. Admin and Super Admin view totals and pending items such as pending approvals and appointment counts.
Alternative / Exception Flows	A1. Attempt to perform an action outside role authority: system rejects and logs the attempt (e.g., Admin editing another Admin).
Postconditions	System remains policy compliant; schedule rules and content moderation are enforced; oversight totals are updated.

APPENDIX A: REFERENCES

Reference	Version/Date	Description	Location
Concept Document Template	Course template	Formatting and section structure	Provided PDF/DOCX
Use Case Examples	Course example	Use case diagram and specification examples used for alignment	Provided PDF
Project Requirements List (Renu)	Course handout	Role capabilities and restrictions used to define actors and use cases	Provided TXT

APPENDIX B: KEY TERMS

The following table provides definitions for terms and acronyms relevant to this concept document.

Term	Definition
Account Approval	Administrative process that determines whether a new applicant is allowed to access the system as an active user.
Availability Window	Date/time range a tutor marks as open for meetings, subject to system-wide limits and holiday restrictions.
Basic Profile	Limited profile information visible across roles (e.g., tutor capability and rating summary) to support scheduling decisions while protecting privacy.
Capability	Course/subject area or tutoring skill that a tutor lists to show what kinds of academic support they provide.
Holiday Block	A date/time range marked unavailable for tutoring sessions by system-level schedule rules.
Meeting Status	State of an appointment: Scheduled, Rescheduled, Canceled, or Completed.
Role-Based Access Control	Security approach where each role is given only the permissions needed to perform allowed tasks.
Review Moderation	Administrative process for removing inappropriate reviews and controlling review visibility according to policy.
Schedule Rules	Global constraints such as allowed tutoring hours and blocked holidays used to validate availability and bookings.
Super Admin	Highest-level oversight role with system-wide visibility; limited from booking appointments or setting tutor availability.